

Plagiarism Detection Benchmark

Dataset

Consolidated overview of the source code plagiarism detection benchmark datasets across Python and Java environments.

Series & Classrooms	Assignment / Benchmark Title	Total	Ground-Truth Breakdown	Detailed Mapping (Student → Classification)
3.1 TS-A Series: Basic Syntax & Lexical Obfuscation <i>(3CSB Python / 4CSA Java)</i>	TS-A 1: Merge Sort (Exact Copy)	2	1 Baseline 1 Type 1 Plagiarized	Mary: MergeSort (Original Baseline) Charles: MergeSortExactCopy (Type 1 Exact Copy of Mary)
	TS-A 2: Quick Sort (Renaming Variables)	2	1 Baseline 1 Type 2 Plagiarized	Mary: QuickSort (Original Baseline) Charles: QuickSortRenamed (Type 2 Renamed of Mary)

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	TS-A 3: Binary Search (Changing Spacing)	2	1 Baseline 1 Type 1 Plagiarized	Mary: BinarySearchTree (Original Baseline) Charles: BinarySearchTreeReformat (Type 1 Spacing/Formatting of Mary)
3.2 TS-B Series: Structural Obfuscation <i>(3CSB Python / 4CSA Java)</i>	TS-B 1: Fibonacci (Swapping Lines)	2	1 Baseline 1 Type 3 Plagiarized	Mary: Fibonacci (Original Baseline) Charles: FibonacciReordered (Type 3 Statement Reordering of Mary)
	TS-B 2: Sorting (Breaking into Functions)	2	1 Baseline 1 Type 3 Plagiarized	Mary: SelectionMonolithic (Original Baseline) Charles: SelectionExtracted (Type 3

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				Method Extraction of Mary)
	TS-B 3: Loops (Swapping Loop Types)	2	1 Baseline 1 Type 3 Plagiarized	Mary: ComputeEvenTotal (Original Baseline - for loop) Charles: EvenNumberSum (Type 3 Control Flow of Mary - while loop)
3.3 TS-C Series: Advanced & Mixed Attacks <i>(3CSB Python / 4CSA Java)</i>	TS-C 1: Quick Sort (Adding Fake Code)	2	1 Baseline 1 Type 3 Plagiarized	Mary: QuickSort (Original Baseline) Charles: QuickSortDead Code (Type 3 Dead Code Injection of Mary)
	TS-C 2: Linked List (Swapping	2	1 Baseline 1 Type 3	Mary: LinkedList

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	Math Logic)		Plagiarized	(Original Baseline) Charles: LinkedListSubstitution (Type 3 Logic Substitution of Mary)
	TS-C 3: Binary Search Tree (Mixed Tricks)	2	1 Baseline 1 Mixed Plagiarized	Mary: BinarySearchTreeOps (Original Baseline) Charles: MixedAttackTree (Mixed Attack: Rename + Reorder + Dead Code)
3.4 Multiple-A Series: Control Groups & Lexical/Format <i>(3CSC Python / 4CSB Java)</i>	Multiple-A 1: Binary Search (Control Group)	10	8 Organic 1 Unique 1 Type 1 Copy	Mary–Patrick (8): Organic / Honest Rachel: Unique Baseline Karo: Type 1 Exact Copy of Rachel

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	Multiple-A 2: Bubble Sort (Renamed & Reformatted)	12	8 Organic 1 Unique 1 Type 1 Exact 1 Type 2 Renamed 1 Type 1 Reformatted	Mary–Patrick (8): Organic / Honest Rachel: Unique Baseline Karo: Type 1 Exact Copy of Rachel Sol: Type 2 Renamed of Rachel Jake: Type 1 Whitespace/Format of Rachel
	Multiple-A 3: 0/1 Knapsack (High-Diversity Control)	14	10 Organic 1 Unique 3 Type 3 Plagiarized	Mary–Karo (10): Organic / Honest Sol: Unique Baseline Ramil: Type 3 Statement Reordering of Mary Alex: Type 3 Method Extraction of Mary Sam: Type 3 Control Flow of

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				Mary
3.5 Multiple-B Series: Multi-Family & Structural Obfuscation <i>(3CSC Python / 4CSB Java)</i>	Multiple-B 1: Linked List Reversal	15	10 Organic 2 Unique Baselines 2 Type 1 Copies 1 Type 3 Combo	Mary–Karo (10): Organic / Honest Sol: Unique Recursive Baseline Ramil: Unique Iterative Baseline Alex: Type 1 Copy of Sol Sam: Type 1 Copy of Ramil Chris: Type 3 Structural + Rename of Ramil
	Multiple-B 2: Binary Tree Traversal	13	9 Organic 1 Unique Baseline 1 Type 1 Copy 2 Type 3 Plagiarized	Mary–Rachel (9): Organic / Honest Karo: Unique Baseline Sol: Type 1 Copy of Karo Ramil: Type 3 Dead Code of Karo

Series & Classrooms	Assignment / Benchmark Title	Total	Ground-Truth Breakdown	Detailed Mapping (Student → Classification)
				Alex: Type 3 Logic Substitution of Karo
3.6 Multiple-C Series: Logic Substitution & Stress Tests <i>(3CSC Python / 4CSB Java)</i>	Multiple-C 1: Two Sum	16	10 Organic 1 Unique Baseline 1 Type 1 Exact 1 Type 2 Renamed 1 Type 3 Structural 2 Mixed Attacks	Mary–Karo (10): Organic / Honest Sol: Unique Baseline Ramil: Type 1 Copy of Sol Alex: Type 2 Renamed of Sol Sam: Type 3 Recursion of Sol Chris: Mixed Attack of Sol Jake: Triple Combo Attack of Sol
	Multiple-C 2: Graph BFS	11	9 Organic 1 Unique Baseline 1 Type 1 Copy	Mary–Rachel (9): Organic / Honest Karo: Unique Baseline

Series & Classrooms	Assignment / Benchmark Title	Total	Ground-Truth Breakdown	Detailed Mapping (Student → Classification)
				Sol: Type 1 Exact Copy of Karo
3.7 Multiple-D Series: 30-Student Cohort Scalability Test <i>(3CSC Python / 4CSB Java)</i>	Multiple-D 1: Quick Sort (Large Cohort)	30	24 Organic 1 Type 1 Exact 2 Type 2 Renamed 2 Type 3 Structural 1 Mixed Attack	Students 1–24: Organic Independent Kiko: Type 1 Exact Copy of Mary Darrel / Edrian: Type 2 Renamed of Mary Sean / Hannah: Type 3 Structural of Mary Stark: Mixed Attack of Mary
3.8 Multiple-E Series: Algorithmic Equivalence & Control Flow Benchmarks <i>(3CSC Python / 4CSB Java)</i>	Multiple-E 1: Grid DFS (Recursive vs. Stack)	10	1 Baseline 2 Organic Variations 2 Type 1 Copies 2 Type 2 Renamed 3 Type 3	Mary: Baseline DFS Charles / Nicolo: Organic Variations (BFS / Set) Dan / Ramon: Type 1 Copies

Series & Classrooms	Assignment / Benchmark Title	Total	Ground-Truth Breakdown	Detailed Mapping (Student → Classification)
			Structural	Jude / Jm: Type 2 Renamed Patrick / Rachel / Karo: Type 3 Structural
	Multiple-E 2: Frequency Counter / Valid Parentheses	10	1 Baseline 2 Organic Variations 2 Type 1 Copies 2 Type 2 Renamed 3 Type 3 Structural	Mary: Baseline Map/Stack Charles / Nicolo: Organic Variations Dan / Ramon: Type 1 Copies Jude / Jm: Type 2 Renamed Patrick / Rachel / Karo: Type 3 Structural
	Multiple-E 3: Range Accumulator / Max Subarray	10	1 Baseline 2 Organic Variations 2 Type 1 Copies 2 Type 2	Mary: Baseline Kadane's Charles / Nicolo: Organic Variations

Series & Classrooms	Assignment / Benchmark Title	Total	Ground-Truth Breakdown	Detailed Mapping (Student → Classification)
			Renamed 3 Type 3 Structural	Dan / Ramon: Type 1 Copies Jude / Jm: Type 2 Renamed Patrick / Rachel / Karo: Type 3 Structural
	Multiple-E 4: Substring Search (Control Flow Obfuscation)	10	1 Baseline 2 Organic Variations 2 Type 1 Copies 2 Type 2 Renamed 3 Type 3 Structural	Mary: Baseline Brute Force Charles / Nicolo: Organic Variations Dan / Ramon: Type 1 Copies Jude / Jm: Type 2 Renamed Patrick / Rachel / Karo: Type 3 Structural
3.9 Multiple-F Series: High-Volume Classroom	Multiple-F 1: Merge Sort (36 Submissions)	36	24 Organic 3 Type 1 Exact 3 Type 2 Renamed	Students 1–24: Organic Independent Students

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Benchmarks <i>(3CSC Python / 4CSB Java)</i>			4 Type 3 Structural 2 Mixed Attacks	25–27: Type 1 Exact Copies of Mary Students 28–30: Type 2 Renamed of Mary Students 31–34: Type 3 Structural of Mary Students 35–36: Mixed Stacked Attacks of Mary
	Multiple-F 2: BST (42 Submissions)	42	28 Organic 4 Type 1 Exact 4 Type 2 Renamed 4 Type 3 Structural 2 Mixed Attacks	Students 1–28: Organic Independent Students 29–32: Type 1 Exact Copies of Mary Students 33–36: Type 2 Renamed of Mary Students 37–40: Type 3 Structural of

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				Mary Students 41–42: Mixed Stacked Attacks of Mary
3.10 Standard DSA Clone Benchmarks <i>(3CSD Python / 4CSC Java)</i>	BST: Clone Benchmark	6	1 Baseline 1 Type 1 2 Type 2 2 Type 3	Mary: Baseline Charles: Type 1 Exact Nicolo / Dan: Type 2 Renamed Ramon / Jude: Type 3 Structural
	Binary Tree: Clone Benchmark	5	1 Baseline 2 Type 2 2 Type 3	Mary: Baseline Charles / Nicolo: Type 2 Renamed Dan / Ramon: Type 3 Structural
	Linked List: Clone Benchmark	5	1 Baseline 1 Type 1 2 Type 2 1 Type 3	Mary: Baseline Charles: Type 1 Exact Nicolo / Dan: Type 2

Series & Classrooms	Assignment / Benchmark Title	Total	Ground-Truth Breakdown	Detailed Mapping (Student → Classification)
				Renamed Ramon: Type 3 Structural
	Merge Sort: Clone Benchmark	6	1 Baseline 1 Type 1 2 Type 2 2 Type 3	Mary: Baseline Charles: Type 1 Exact Nicolo / Dan: Type 2 Renamed Ramon / Jude: Type 3 Structural
	Quick Sort: Clone Benchmark	6	1 Baseline 1 Type 1 2 Type 2 2 Type 3	Mary: Baseline Charles: Type 1 Exact Nicolo / Dan: Type 2 Renamed Ramon / Jude: Type 3 Structural